1. recall that coal is mainly carbon;
2. recall that petrol, diesel fuel and fuel oil are mainly compounds of hydrogen and carbon (hydrocarbons);
3. recall that, when fuels burn, atoms of carbon and/or hydrogen from the fuel combine with atoms of oxygen from the air to produce carbon dioxide and/or water (hydrogen oxide);
4. recall that atoms are rearranged during a chemical reaction;
5. interpret representations of the rearrangement of atoms during a chemical reaction;
6. understand that during the course of a chemical reaction the numbers of atoms of each element must be the same in the products as in the reactants;
7. understand that the conservation of atoms during combustion reactions has implications for air quality;
8. recall that the properties of the reactants and products are different;
9. understand how sulfur dioxide is produced if the fuel contains any sulfur;
10. understand how burning fossil fuels in power stations and for transport pollutes the atmosphere with:
    - carbon dioxide and sulfur dioxide,
    - carbon monoxide and particulate carbon (from incomplete burning),
    - nitrogen oxides (from the reaction between atmospheric nitrogen and oxygen at the high temperatures inside engines);
11. relate the formulas for carbon dioxide CO_2, carbon monoxide CO, sulfur dioxide SO_2, nitrogen monoxide NO, nitrogen dioxide, NO_2 and water H_2O, to visual representations of their molecules;
12. **recall that nitrogen monoxide NO, is formed during the combustion of fuels in air, and is subsequently oxidised to nitrogen dioxide NO_2 (NO and NO_2 are jointly referred to as 'NOx');**
13. understand that atmospheric pollutants cannot just disappear, they have to go somewhere:
    - particulate carbon is deposited on surfaces, making them dirty;
    - sulfur dioxide and nitrogen dioxide react with water and oxygen to produce acid rain;
    - carbon dioxide is used by plants in photosynthesis;
    - carbon dioxide dissolves in rain water and in sea water.

ⓘ Candidates are not required to write word or symbol equations.